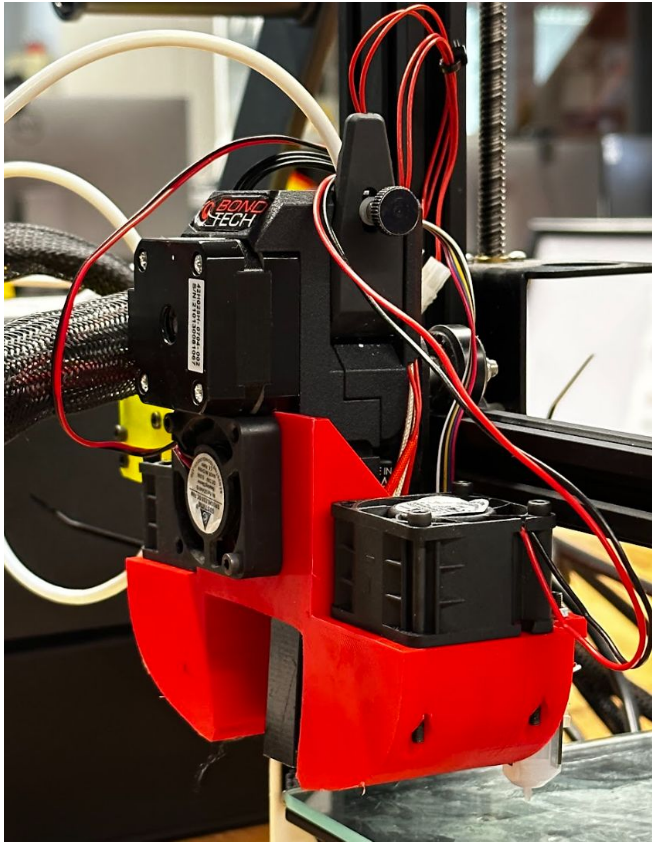
HOTEND PROJECT MONASH CONSULTING

Team 11

Jason Abi Chebli, Josh Audric Tobias, Nicholas Phouthasenh, Daniel Kriekhoff, Malcolm Mill

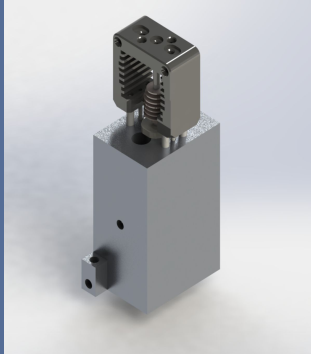


ACKNOWLEDGEMENT OF COUNTRY

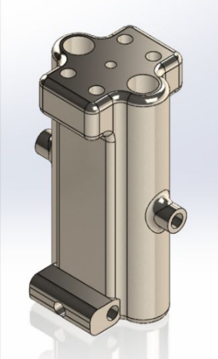
Karkarook Lake in Karkarook Park

Karkarook means "Sandy Place" in the Bunurong language

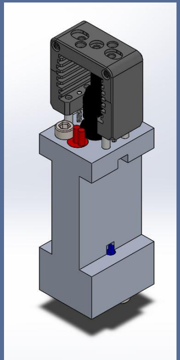
CONSOLIDATION OF CONCEPTS



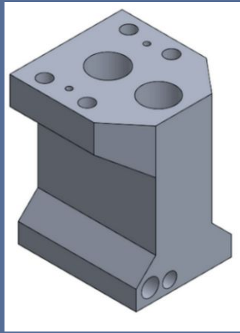
Team 11



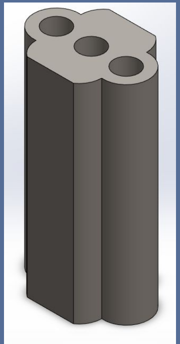
Team 12



Team 13



Team 14



Team 15

Final Heat Block

FINAL HOTEND DESIGN

60W Heater Cartridge x2

Slice Mosquito Heat Break

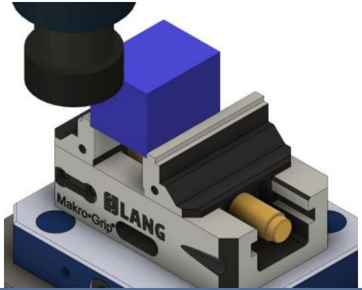
Slice Mosquito Heatsink

SuperVolcano Nozzle

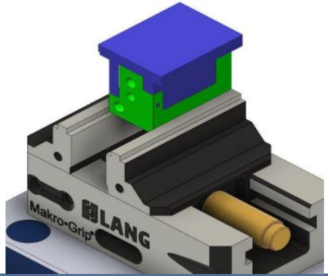
PT1000 Thermistor

Slice Mosquito Standoff

FUSION 360 CAM (5-AXIS)

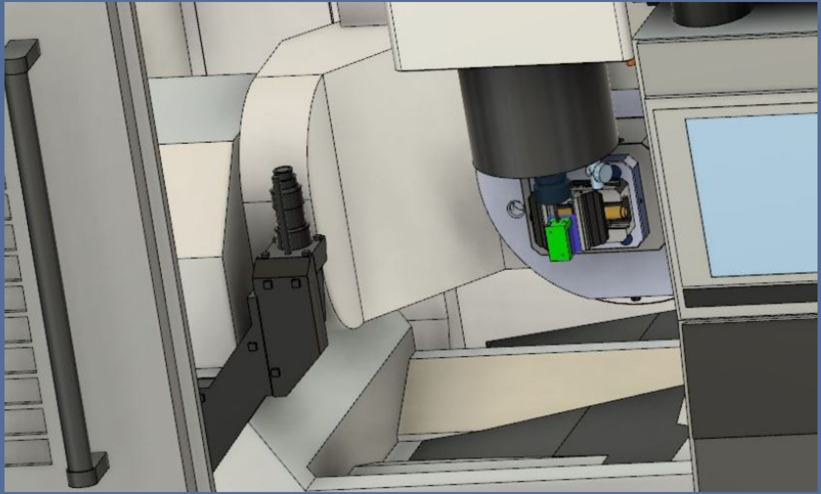


SETUP 1



SETUP 2

OKUMA GENOS M460V-5AX



TOTAL CYCLE TIME = 2 min 43 s

TOTAL MACHINING TIME = 25 min

MATERIAL SELECTION

MATERIALS

Aluminium (6061)

Stainless Steel (316)

Brass (CZ126)

Copper Alloy (110)

Titanium (Grade 12)

PROPERTIES

Property	***Weight***
Thermal Conductivity (W/mK)	10
Machinability Rating (%)	8
Price (AUD/100mm)	7
CO_2 Emissions (tonnes/product produced)	5
Yield Strength (MPa)	5
Density (kg/m^3)	3
Thermal Expansion Coefficient (1/degree)	2

Material	Aluminium (6061)	Stainless Steel (316)	Copper Alloy (110)	Titanium (Grade 12)	Brass (CZ126)
Total (/400)	**327**	233	212	147	190

THERMAL AND STRUCTURAL ANALYSIS

SLICE MOSQUITO HOTEND

E3D HOTEND

FINAL HOTEND DESIGN

Thermal Analysis

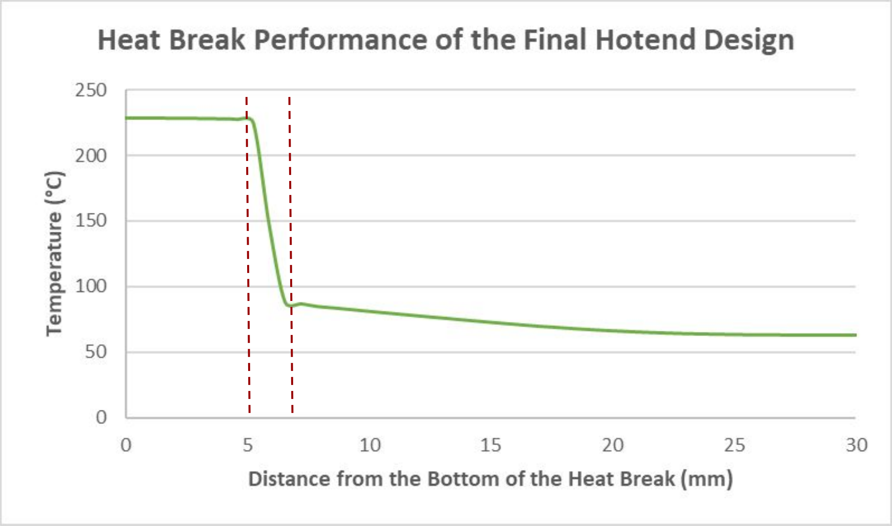
- *Heat Break Performance*
- *Heat Block Performance*
- *PLA Flow Rate Performance*

Structural Analysis

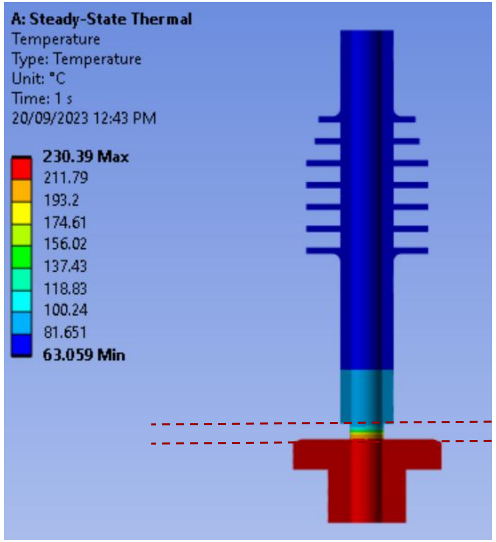
- *X, Y and Z -hit*
- *Back Pressure*
- *1 and 2 Hand Torque*

THERMAL ANALYSIS

RESULTS - HEAT BREAK



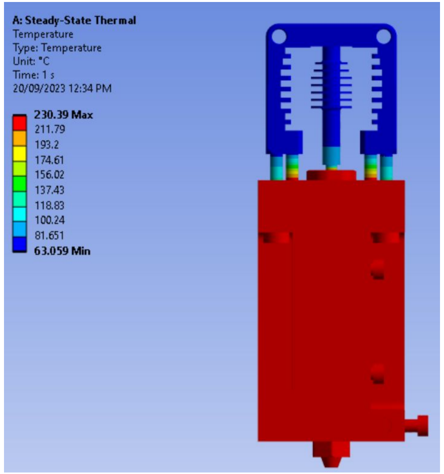
HEAT BREAK PERFORMANCE

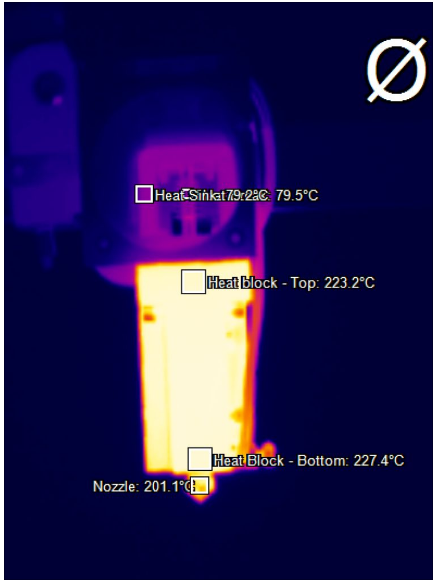


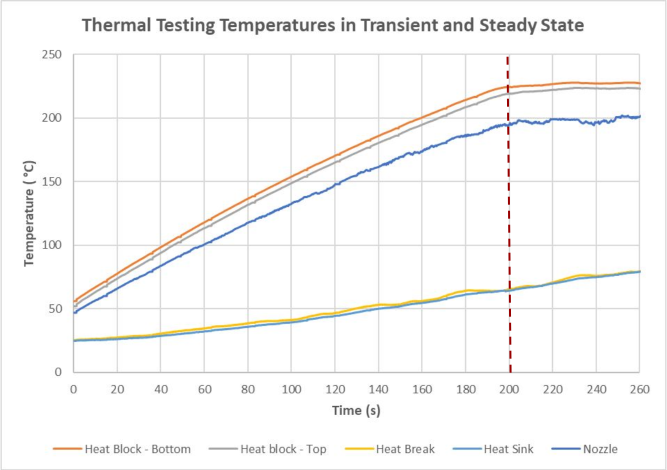
HEAT BREAK PERFORMANCE - ANSYS

THERMAL ANALYSIS

RESULTS - HEAT BLOCK







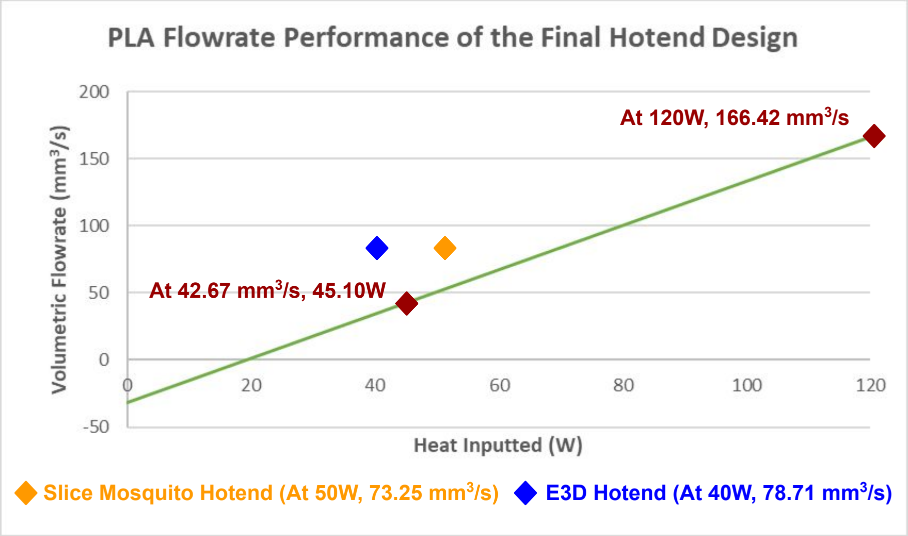
HEAT BLOCK PERFORMANCE - ANSYS

HEAT BLOCK - THERMAL IMAGING

THERMAL IMAGING RESULTS

THERMAL ANALYSIS

RESULTS - PLA FLOW RATE



At 42.67 mm^3/s:

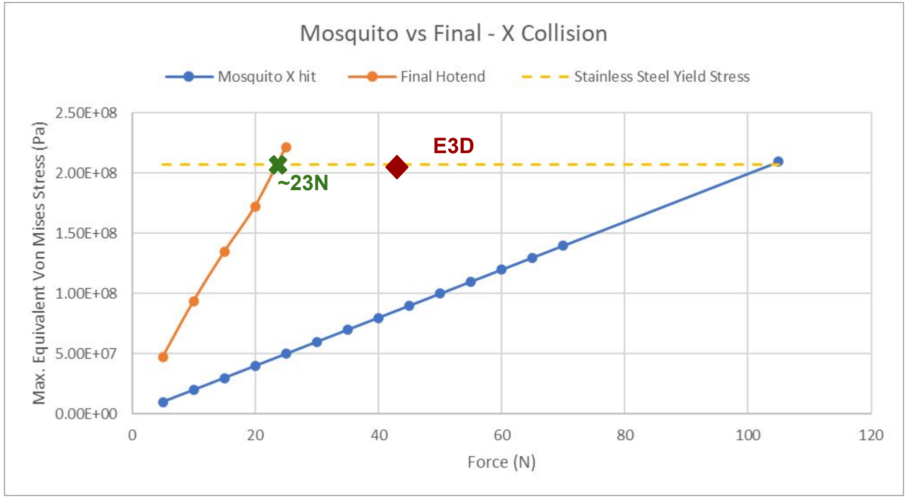
Heat Lost	*Heat Transfer (W)*
Natural Convection	9.47 (21.0%)
Forced Convection	4.25 (9.4%)
Radiation	5.55 (12.3%)
Filament	25.83 (57.3%)
Total Power Input (W)	**45.10**
Duty Cycle (%)	**37.58**

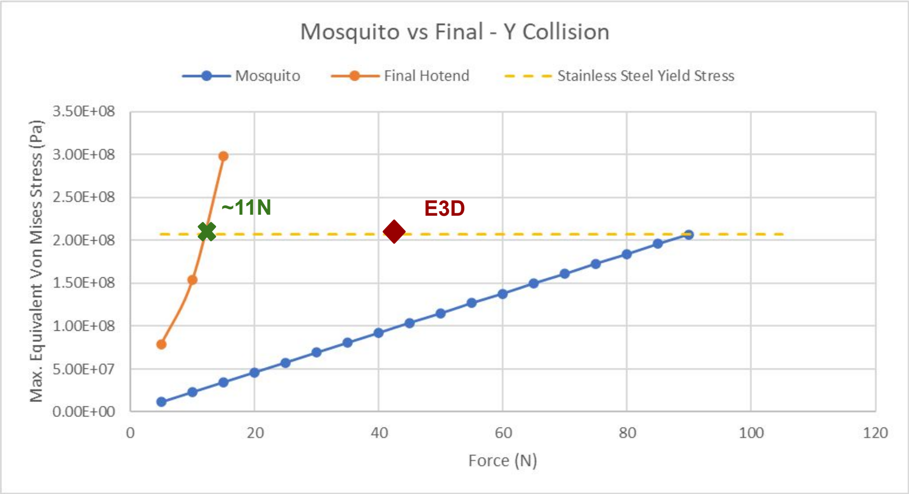
PLA FLOW RATE PERFORMANCE

POWER INPUT, OUTPUT AND DUTY CYCLE

STRUCTURAL ANALYSIS

RESULTS - X, Y HITS



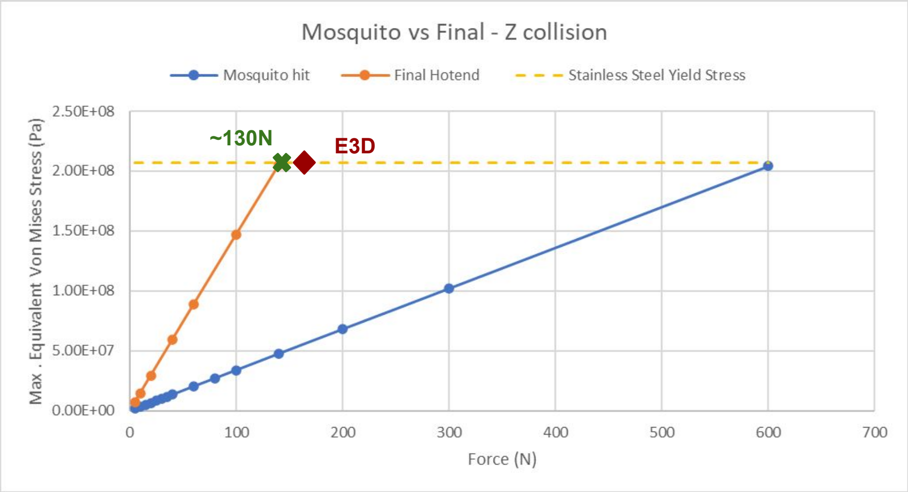


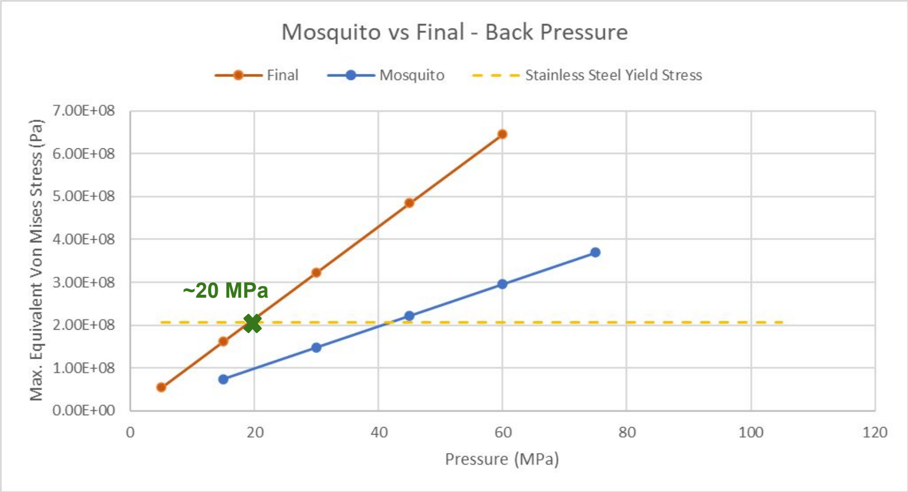
X-HIT

Y-HIT

STRUCTURAL ANALYSIS

RESULTS - Z HIT AND BACK PRESSURE



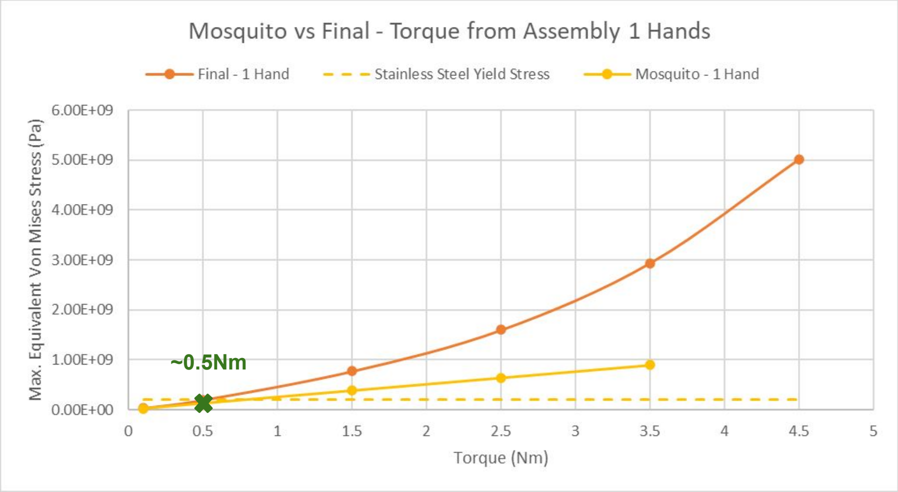


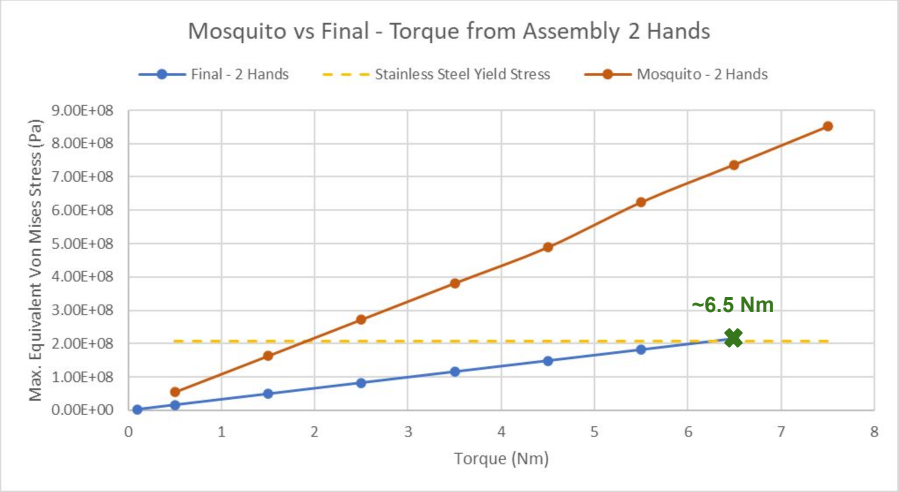
Z-HIT

BACK PRESSURE

STRUCTURAL ANALYSIS

RESULTS - 1 AND 2 HAND TORQUE

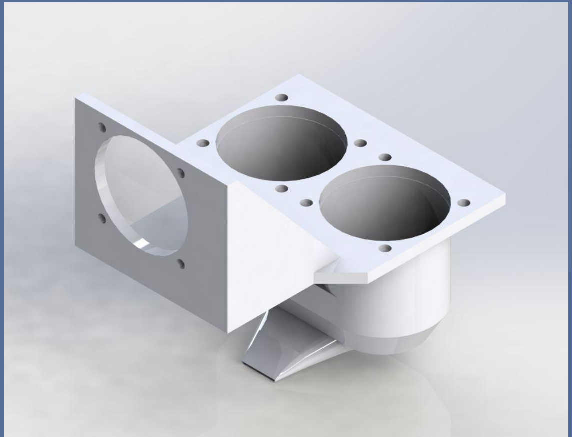


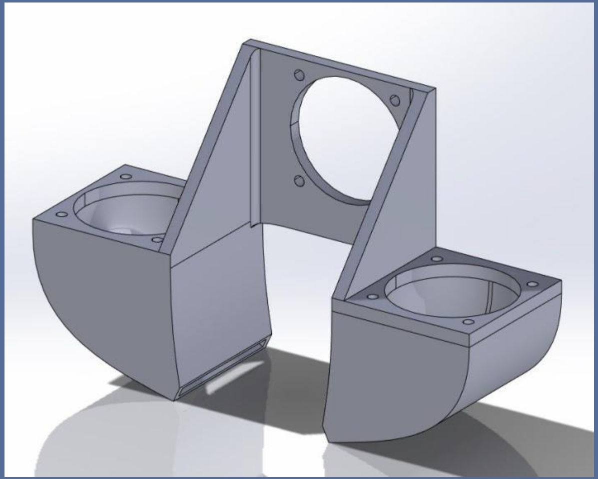


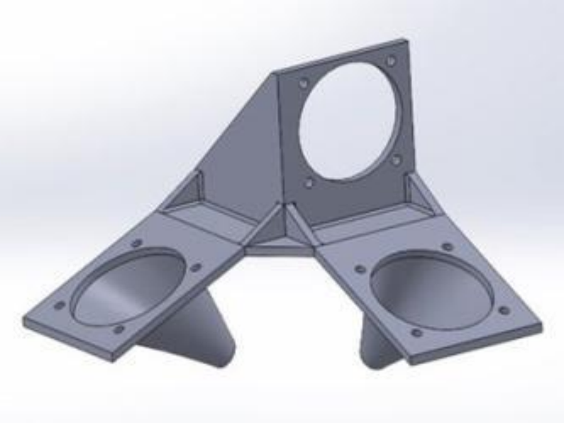
ONE HAND TORQUE

TWO HANDS TORQUE

CFD ANALYSIS







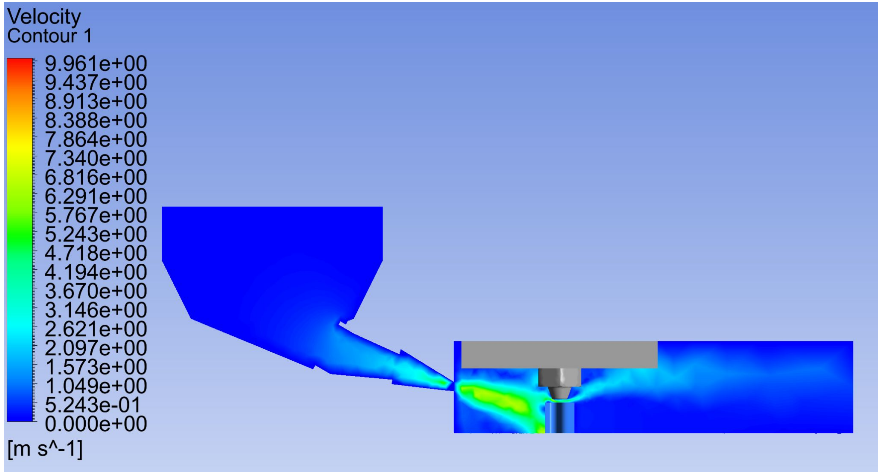
DESIGN 1

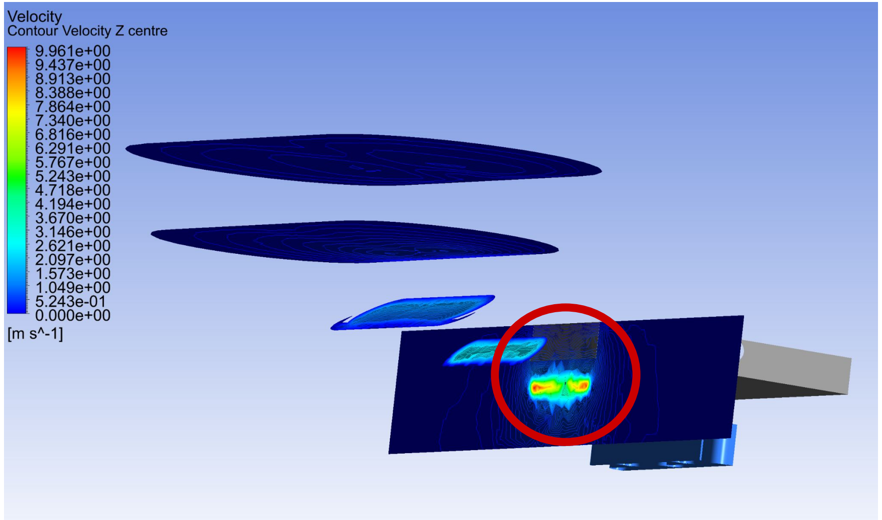
DESIGN 2

DESIGN 3

CFD ANALYSIS

RESULTS FROM DESIGN 1



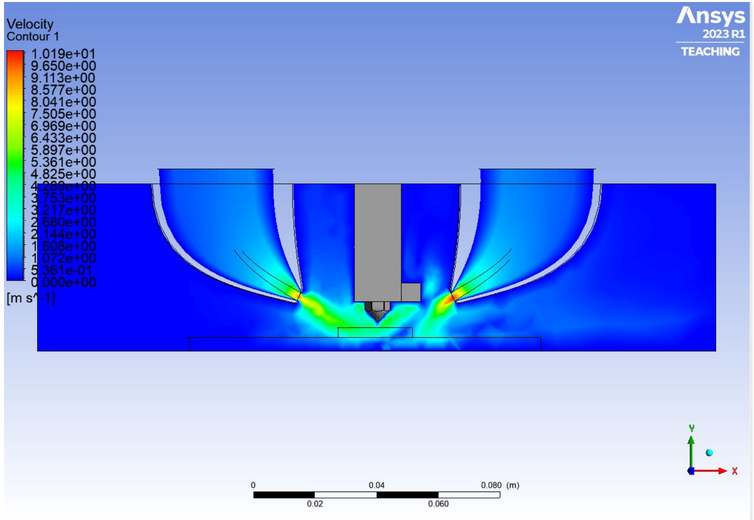


Velocity contour parallel to direction of airflow

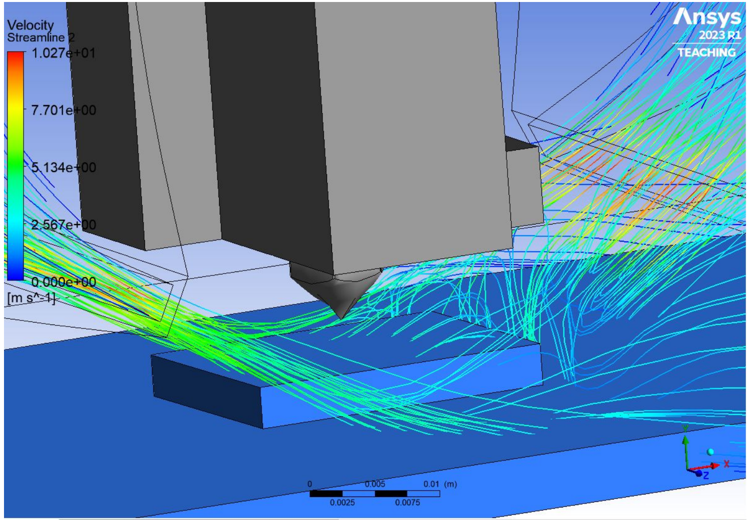
Velocity contour through duct

CFD ANALYSIS

RESULTS FROM DESIGN 2



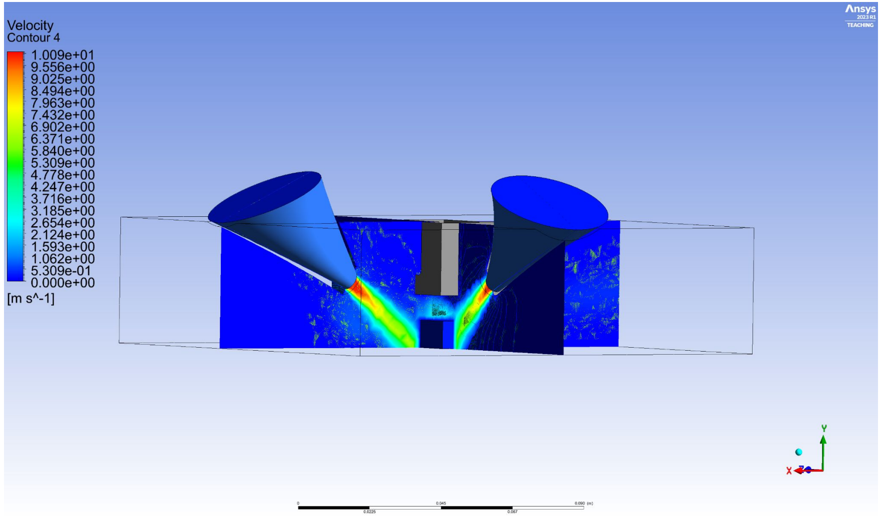
Velocity contour parallel to direction of airflow



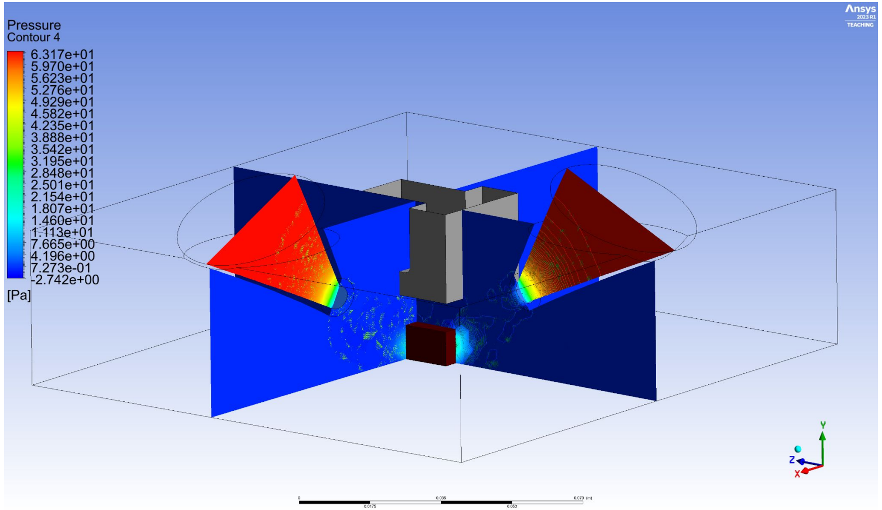
Velocity streamlines close up

CFD ANALYSIS

RESULTS FROM DESIGN 3

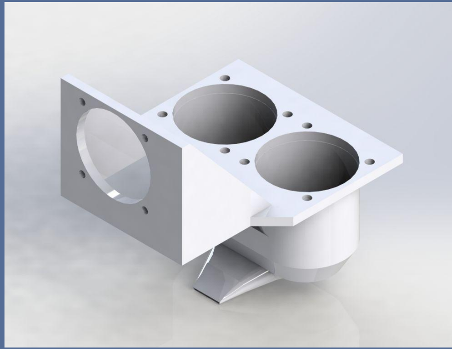


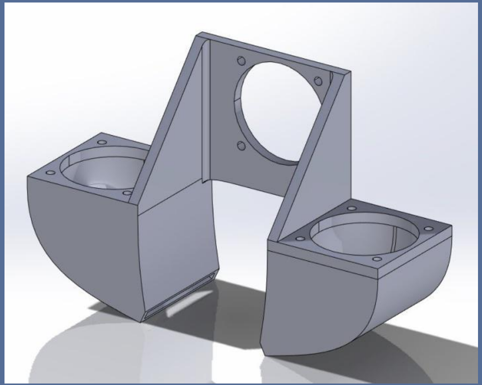
Velocity contours of planes parallel to the airflow

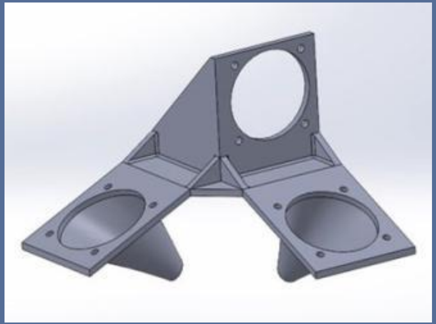


Pressure contours of planes parallel to the airflow

CFD ANALYSIS

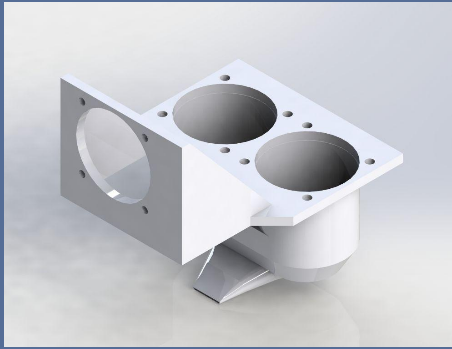


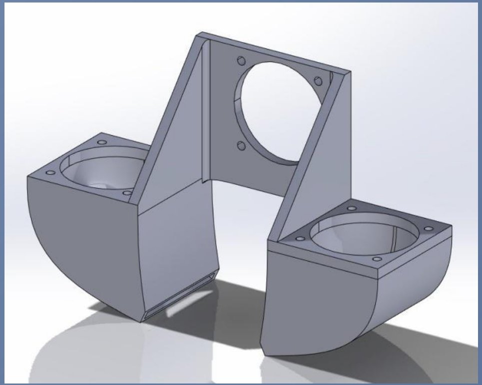


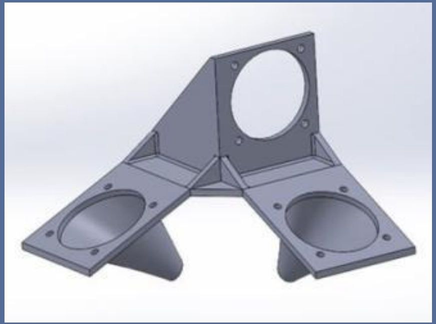


PROS	- Great orientation	- High surface area contact	- High velocity concentration
CONS	- Lack of velocity increase - Limited surface area contact	- Poor orientation of ducts	- Poor orientation

CFD ANALYSIS



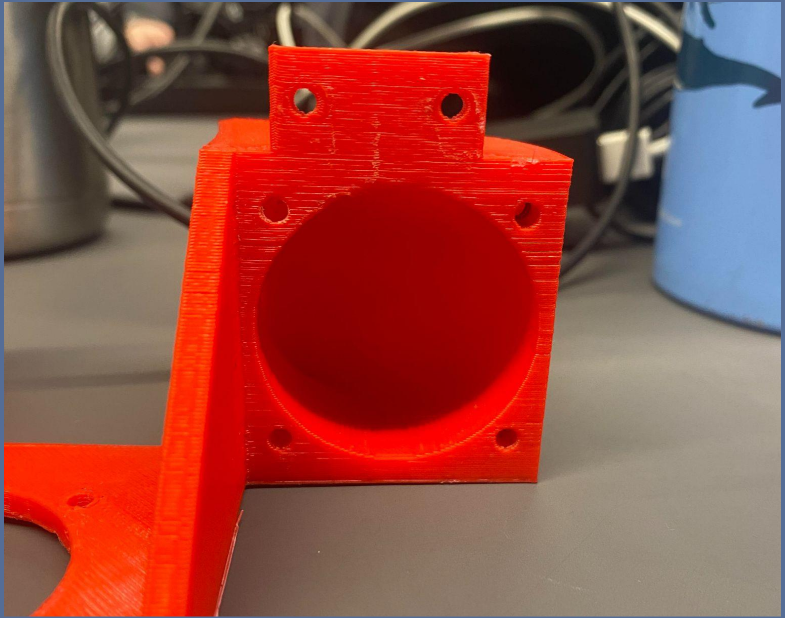




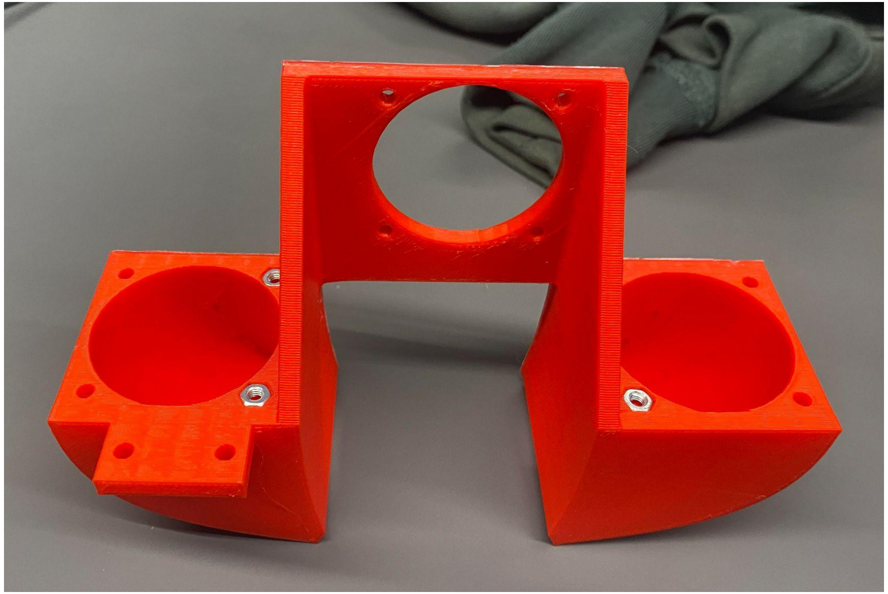
PROS	- Great orientation	- High surface area contact	- High velocity concentration
CONS	- Lack of velocity increase - Limited surface area contact	- Poor orientation of ducts	- Poor orientation

FAN DUCT MANUFACTURING

OFF THE PRINTING BED



NUTS GLUED INTO MOUNTING POINTS



FAN DUCT MANUFACTURING

FINISHED PRODUCT ON ENDER 3

BACK SIDE WITH BL TOUCH SENSOR

FLOW RATE TESTING

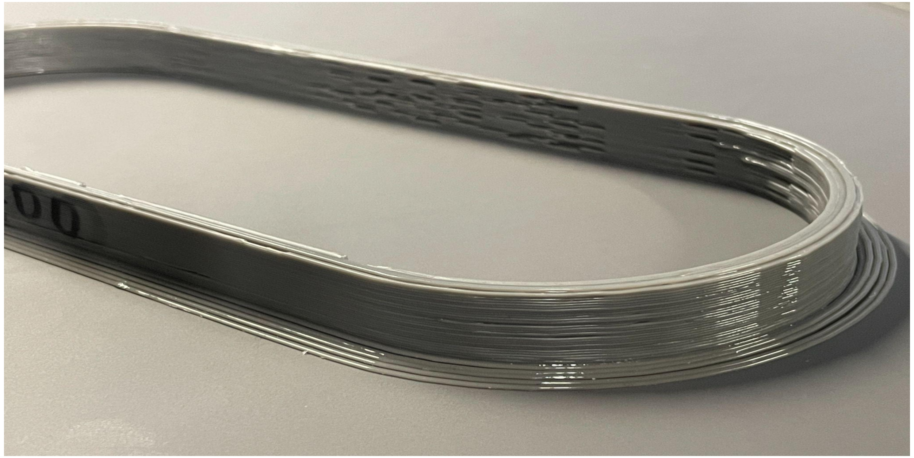
METHOD 1

Slice Engineering Model

RESULTS	
Experimental Flow Rate (mm^3/s)	24.31
Printing Speed (mm/s)	90-100

FLOW RATE TESTING

METHOD 2

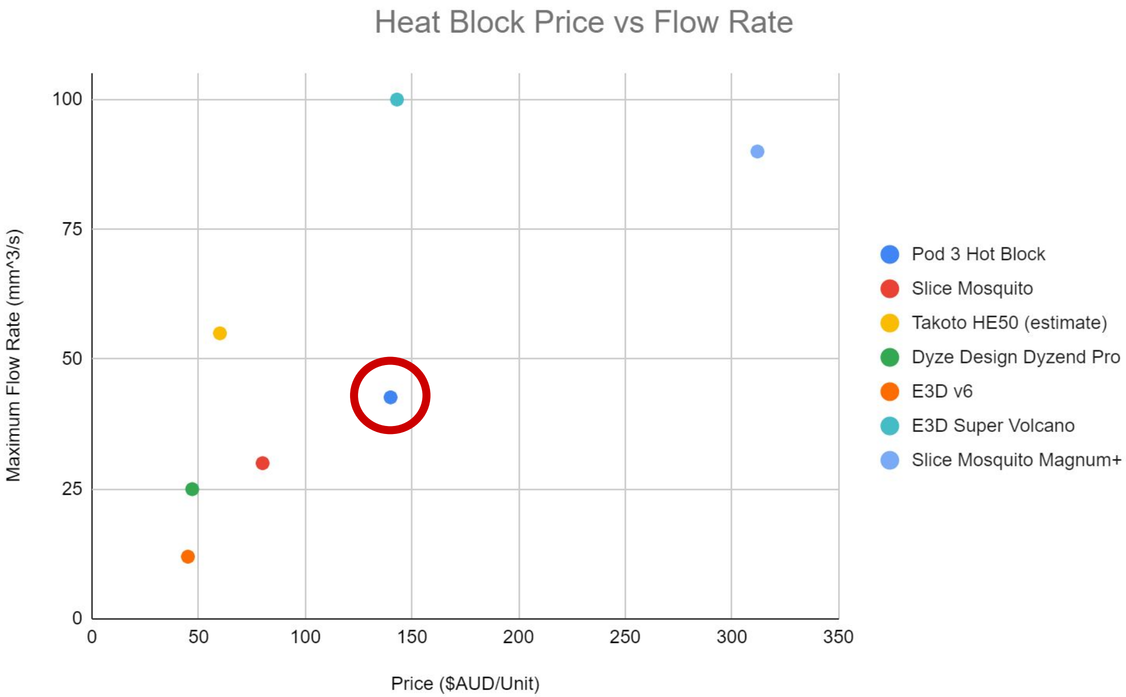


Rack Track Model

RESULTS	
Experimental Flow Rate (mm^3/s)	42.67
Printing Speed (mm/s)	160

FLOW RATE TESTING

MARKET EVALUATION

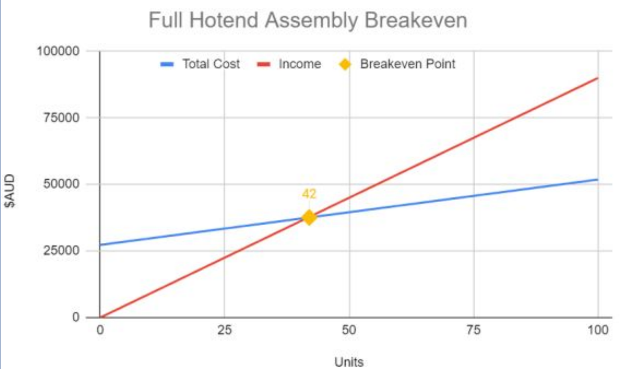


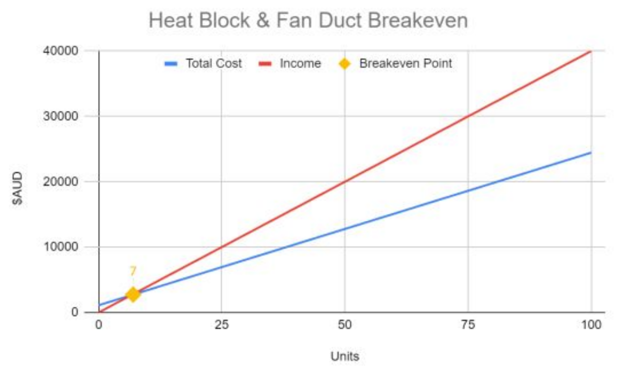
RESULTS	
Experimental Flow Rate (mm^3/s)	17
Printing Speed (mm/s)	160
Cost ($AUD)	140

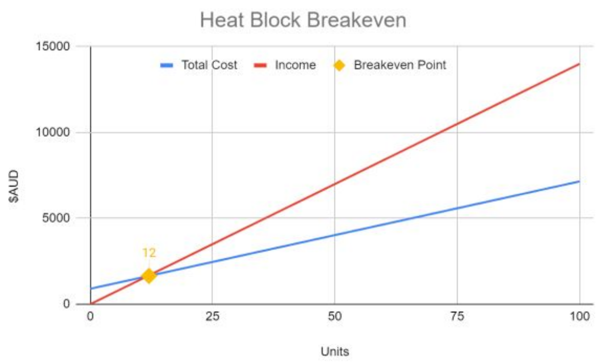
COSTING AND RECOMMENDED RETAIL PRICE

FOR 100 UNITS

Products	***Total Fixed Cost (AUD)***	***RRP per Unit excluding GST (AUD)***	***Profit Margin (%)***
Full Assembly	27,297.50	900.00	42.39
Heat Block and Fan Duct Package	1,102.50	400.00	38.37
Heat Block Only	900.00	140.00	48.93



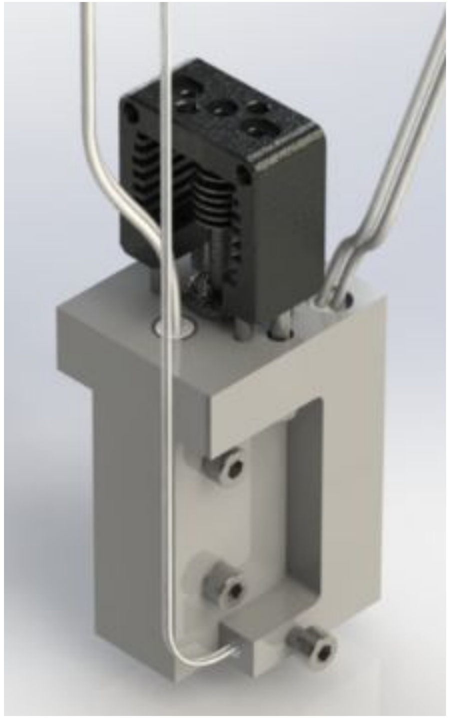




CONCLUSION & RECOMMENDATIONS

RECOMMENDATIONS

1. CONDUCT A MACHINING SIMULATION
2. INVESTIGATE THE USE OF A SINGLE HEATER CARTRIDGE
3. CONSIDER SELLING JUST THE HEAT BLOCK
4. RUN A PRINT SPEED OF 160 MM/S

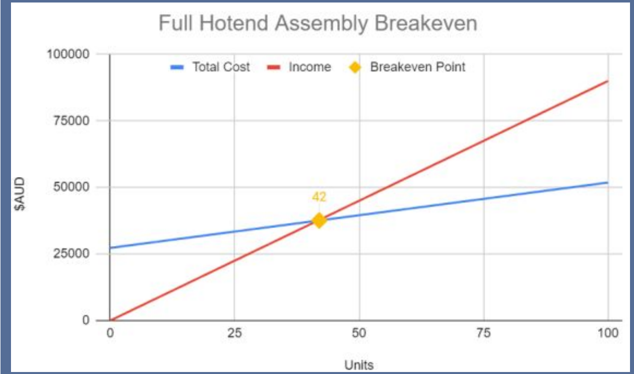


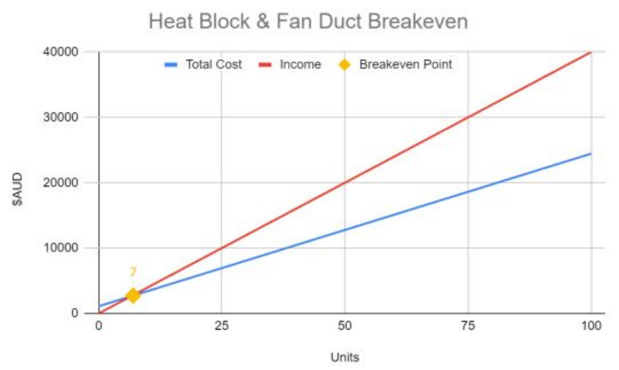
THANKS

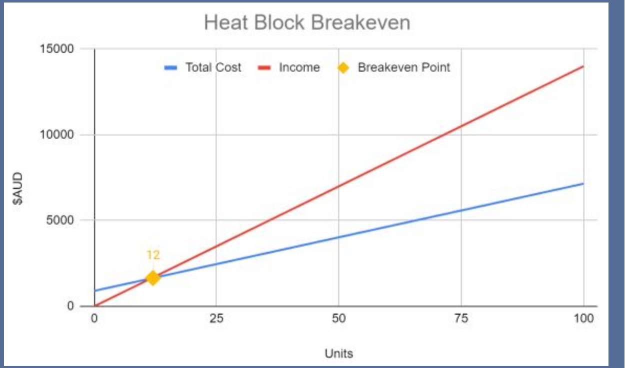
Does anyone have any questions?

Monash University
ABN 12 377 614 012
Wellington Road, Clayton, Victoria Australia 3800
www.monash.edu/

COSTING AND RECOMMENDED RETAIL PRICE





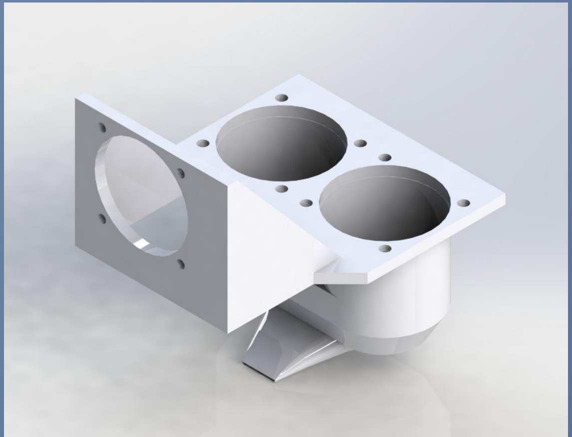


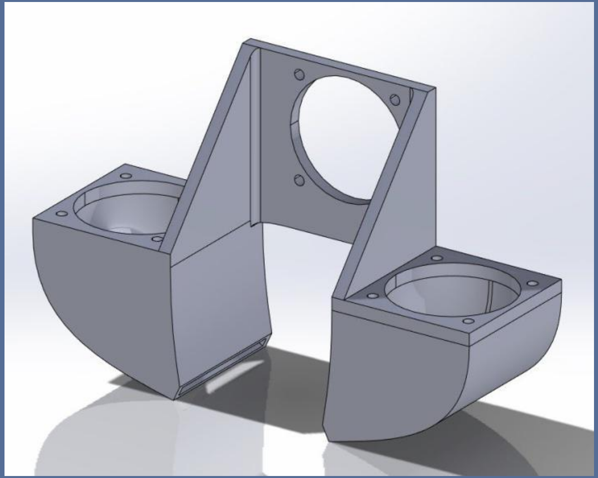
FULL HOTEND ASSEMBLY

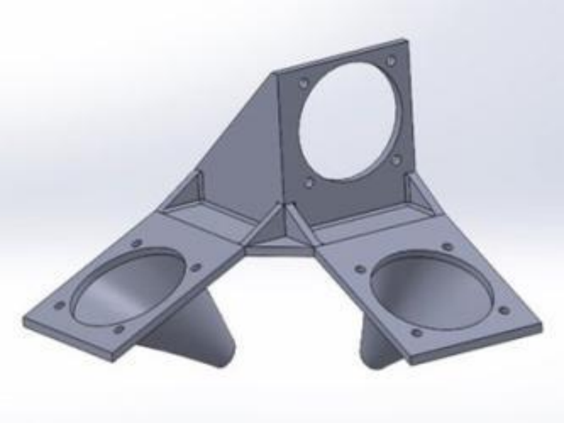
HEAT BLOCK AND FAN DUCT

HEAT BLOCK

CFD ANALYSIS





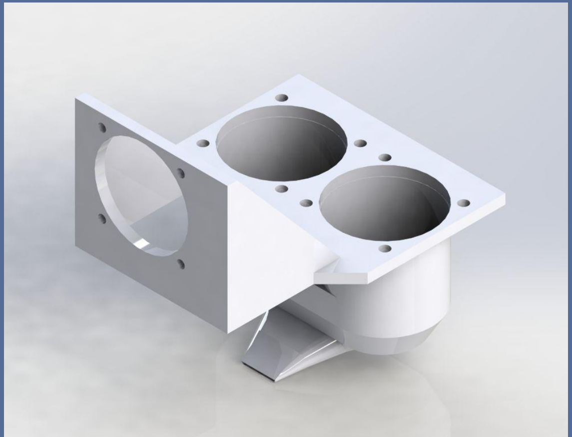


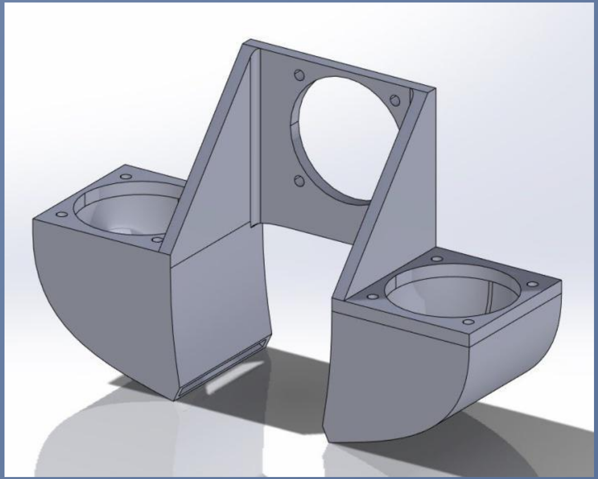
DESIGN 1

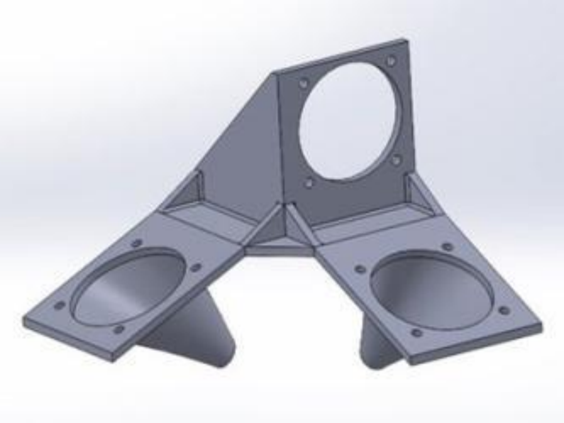
DESIGN 2

DESIGN 3

CFD ANALYSIS





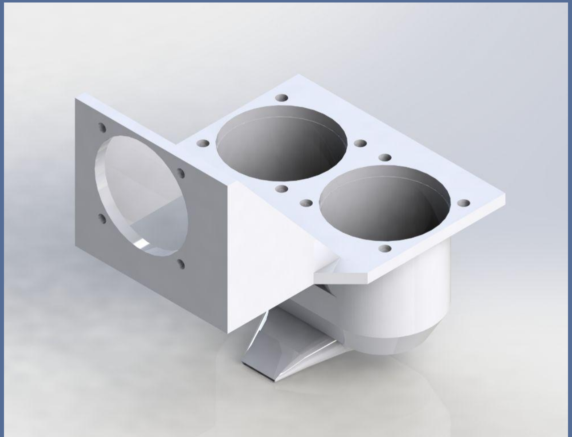


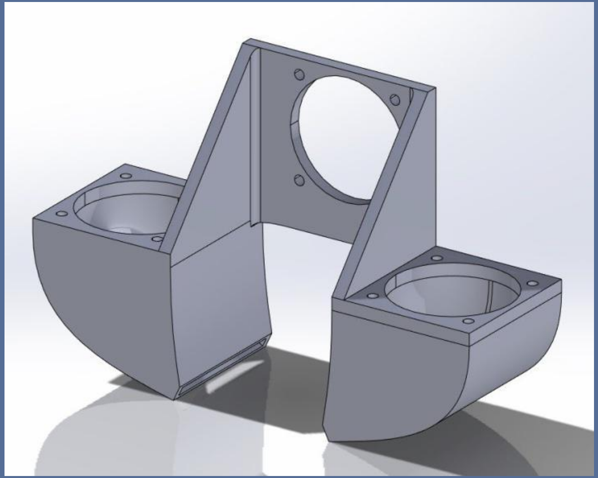
DESIGN 1

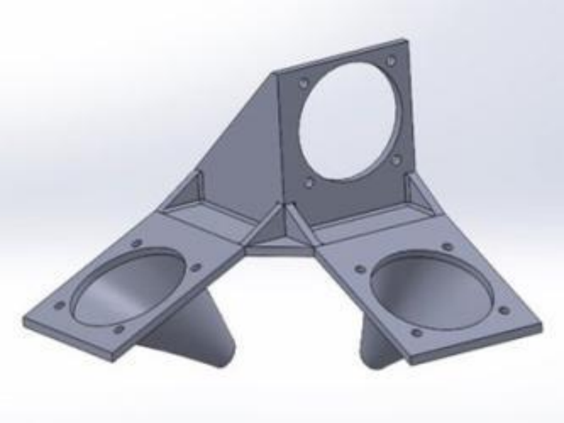
DESIGN 2

DESIGN 3

CFD ANALYSIS







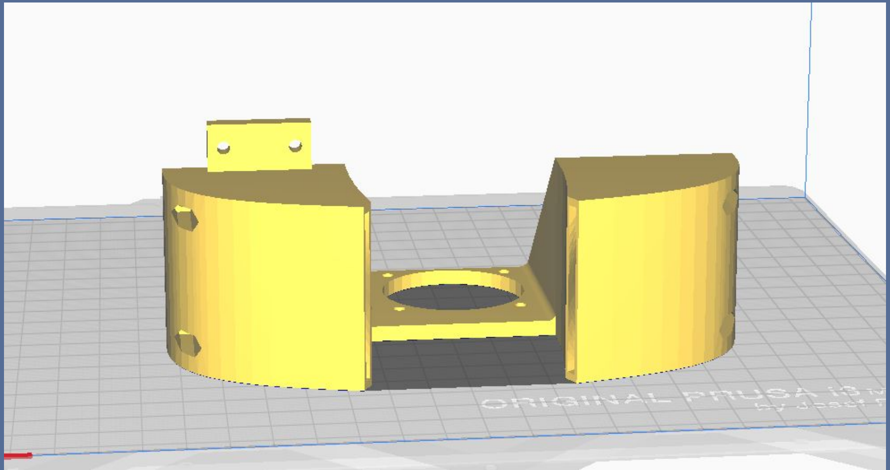
DESIGN 1

DESIGN 2

DESIGN 3

FAN DUCT MANUFACTURING

CURA SLICING BED

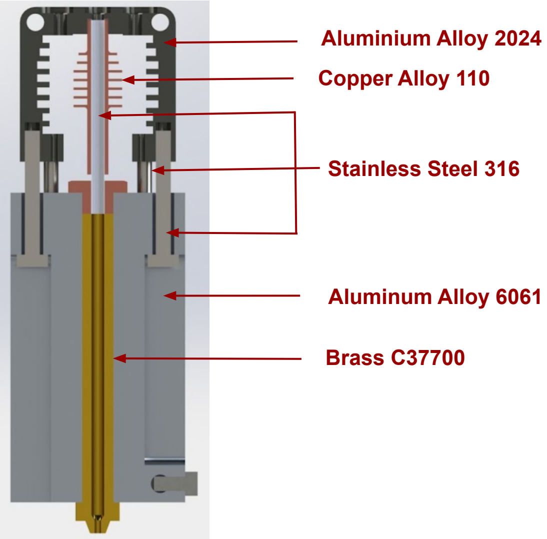


SLICE SETTINGS

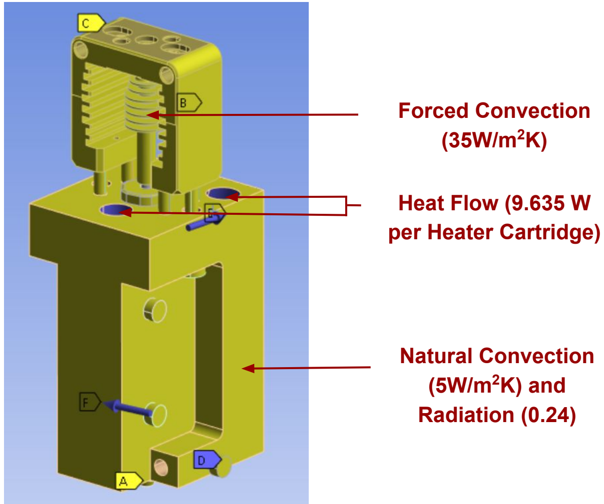
Settings	
Infill (%)	17
Print Speed (mm/s)	60
Layer Height (mm)	0.2
Print Temperature (℃)	200
Build Plate Temperature (℃)	60
Supports	None
Build Plate Adhesion Type	Brim

THERMAL ANALYSIS

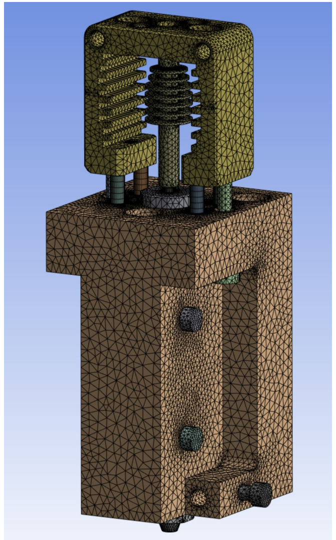
SETUP



MATERIAL ASSIGNMENT



ANSYS LOAD CASES



MESH GENERATED

STRUCTURAL ANALYSIS

RESULTS

E3D Safety Factor for X hit at 40N.

Time [s]	*Minimum*	*Maximum*	*Average*
1	1.489	15	14.709

E3D Safety Factor Y hit 40N.

Time [s]	*Minimum*	*Maximum*	*Average*
1	1.335	15.	14.7

E3D Safety Factor for Z hit 40N.

Time [s]	*Minimum*	*Maximum*	*Average*
1	6.4684	15.	14.956

CONTRACT OF WORK

DESCRIPTION	QUANTITY	UNIT PRICE	TOTAL
LABOUR COST			
Engineering Graduates 40 hours/week for 5 weeks (including Superannuation)	5	$63.08 /hr	$88,312.00
Senior Engineer Supervisor 20 hours/week for 5 weeks (including Superannuation)	1	$209.05 / hr	$29,267.00
Machinist 1 hours/heat block for 2 heat block iterations (including Superannuation)	1	$ 52.46 / hr	$52.46
		TOTAL LABOUR COST	$117,631.46

OVERHEAD COST			
Office Space for the whole team for the 5 weeks	1	$26,849.30	$26,849.30
Utilities for the whole team for the 5 weeks including Electricity and Gas, Internet, Water and Office Kitchen	1	$11,595.41	$11,595.41
Administration Expenses for the whole team for the 5 weeks such as paying wages and salaries	1	$14,998.02	$14,998.02
Security Expenses for the whole team for the 5 weeks	1	$27,000.00	$27,000.00
Maintenance and Cleaning Experiences for the whole team for the 5 weeks	1	$7,000.00	$7,000.00
		TOTAL OVERHEAD COST	$87,442.73

MATERIAL COST			
Slice Mosquito Heat Sink	1 (+1 Spare)	$63.00	$126.00
Slice Mosquito Heat Break - 1.75 mm	1 (+1 Spare)	$64.00	$128.00
Slice Mosquito Heat Block Hardware including Standoff tubes, Bolts and other Bolts not included by Slice	1 (+1 Spare)	$15.00	$30.00
60 W Heater Cartridge	2 (+1 Spare)	$55.00	$165.00
N052AR Fan Duct Fans	2 (+1 Spare)	$20.00	$40.00
GDA010 Heat Break Fan	1 (+1 Spare)	$16.95	$33.90
BLTouch: Auto Bed Leveling Sensor	1 (+1 Spare)	$85.70	$171.40
PT1000 Thermistor	1 (+1 Spare)	$30.00	$120.00
Brass E3D Super Volcano Nozzle	1 (+1 Spare)	$16.00	$32.00
Bondtech DDX Extruder	1(+1 spare)	$179.95	$359.90
Aluminium Alloy 6061 T6 40x40mm Square	1	$107.90	$107.90
bar 1m Length			
PLA 3D Filament 1kg Roll	2 (+1 spare)	$34.95	$104.85
		SUB TOTAL MATERIAL COST	$1,438.95
		SHIPPING COST PERCENTAGE ESTIMATE	30%
		SHIPPING COST ESTIMATE	$431.69
		TOTAL MATERIAL COST	$1,870.63

CONTRACT OF WORK

TECHNOLOGY AND TOOLS COST			
ANSYS Software Licence for the Whole Team for the 7 Weeks for the following licences: Ansys Mechanical, Ansys Mechanical Solver, Ansys Structural, Ansys Structural Solver, Ansys Mechanical CFD-Flo	1	$12,197.26	$12,197.26
SolidWorks Premium Software Licence for the Whole Team for the 5 Weeks	1	$5,747.67	$5,747.67
Fusion360 Software Licence for the Whole Team for the 5 Weeks	1	$525.00	$525.00
MATLAB Software Licence for the Whole Team for the 5 Weeks	1	$782.47	$782.47
Microsoft 365 Business Premium Software Licence for the Whole Team for the 5 Weeks	1	$226.50	$226.50
Zoom Business Plus for the Whole Team for the 5 Weeks	1	$143.83	$143.83
Thermal Camera testing per hour	6	$8.00	$48.00
Prusa 3D Printer fan duct manufacturing per hour	10	$4.00	$40.00
CR-10 3D Printer flow rate testing per hour	15	$2.18	$32.69
		TOTAL TECHNOLOGY AND TOOLS COST	$19,743.42

MISCELLANEOUS COSTS			
Intellectual Property - Design Rights Registration	1	$450.00	$450.00
Business Insurance Costs	1	$262.50	$262.50
Legal Costs	1	$1,925.00	$1,925.00
		MISCELLANEOUS COST	$2,637.50

Quote Total (including GST) = $298,123.46